

Experiment – 20

CSA1717 - Artificial Intelligence

Name: Hemalakshmi H

Reg. No.: 192571049

Aim:

To write a Python program to implement Breadth-First Search (BFS) traversal for a graph.

Algorithm:

1. Start the program.
2. Create a graph using an adjacency list.
3. Initialize an empty queue and a visited set.
4. Add the starting node to the queue and mark it as visited.
5. While the queue is not empty:
 - Remove the front node from the queue.
 - Display the node.
 - Add all unvisited neighbors of the node to the queue.
6. Stop the program.

Source code:

```
# Program to implement Breadth-First Search (BFS)
```

```
from collections import deque
```

```
def bfs(graph, start):
```

```
    visited = set()
```

```
    queue = deque([start])
```

```
    visited.add(start)
```

```
    print("BFS Traversal:", end=" ")
```

```
while queue:
```

```
    node = queue.popleft()
```

```
    print(node, end=" ")
```

```
    for neighbor in graph[node]:
```

```
        if neighbor not in visited:
```

```
            visited.add(neighbor)
```

```
            queue.append(neighbor)
```

```
# Graph representation
```

```
graph = {
```

```
    'A': ['B', 'C'],
```

```
    'B': ['D', 'E'],
```

```
    'C': ['F'],
```

```
    'D': [],
```

```
    'E': ['F'],
```

```
    'F': []
```

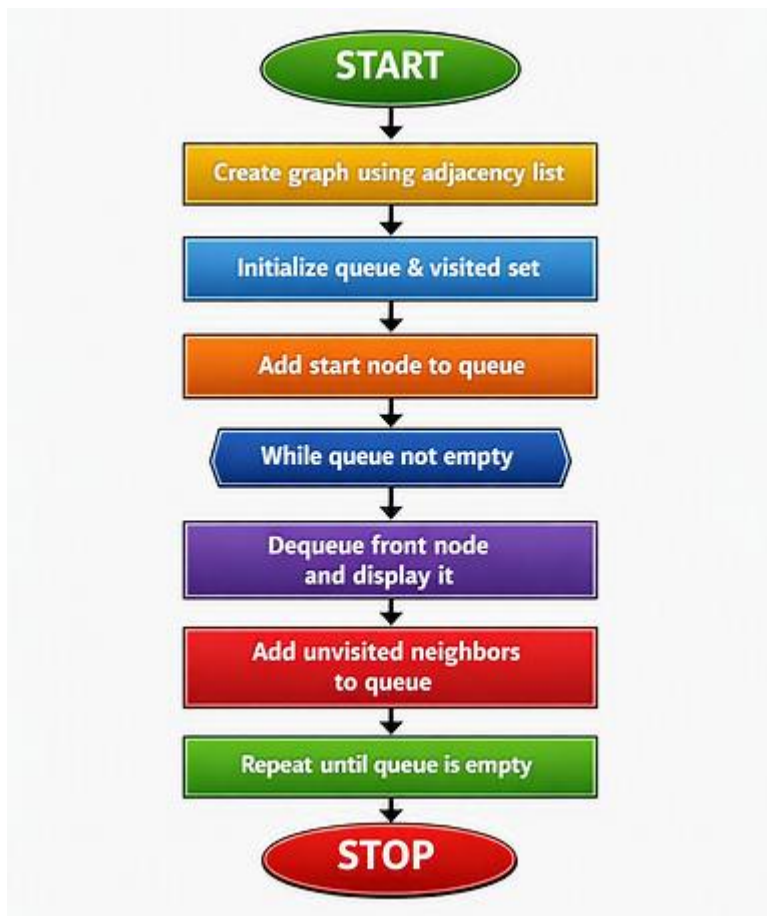
```
}
```

```
# Starting node
```

```
start_node = 'A'
```

```
bfs(graph, start_node)
```

Flowchart:



Output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/hemal/OneDrive/Documents/SIMATS/AI/AI_PY/EXP 20_code_AI.py =
BFS Traversal: A B C D E F
```

Result:

The program successfully implements Breadth-First Search (BFS) traversal for a graph using a queue data structure.